

MACHINE LEARNING ALGORITHMS

NOTE:

The algorithms listed under headings A, B, and C are organized according to students' respective serial numbers.

Student(s)

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A. Supervised Learning Algorithms

01. Linear Regression
02. Logistic Regression
03. Support Vector Machines (SVM)
04. Decision Trees
05. Random Forests
06. Gradient Boosting Machines (GBM)
07. AdaBoost
08. k-Nearest Neighbors (k-NN)
09. Naive Bayes
10. Neural Networks
11. Elastic Net
12. Ridge Regression
13. Lasso Regression
14. XGBoost
15. LightGBM
16. CatBoost
17. Quadratic Discriminant Analysis (QDA)
18. Multinomial Naive Bayes

B. Unsupervised Learning Algorithms

01. K-Means Clustering
02. Hierarchical Clustering

03. DBSCAN (Density-Based Spatial Clustering of Applications with Noise)
04. Principal Component Analysis (PCA)
05. t-Distributed Stochastic Neighbor Embedding (t-SNE)
06. Autoencoders
07. Gaussian Mixture Models (GMM)
08. Isolation Forest
09. Factor Analysis
10. Self-Organizing Maps (SOM)
11. Affinity Propagation
12. Latent Dirichlet Allocation (LDA)
13. Singular Value Decomposition (SVD)
14. Kernel PCA
15. Fuzzy C-Means Clustering
16. Mean Shift
17. Non-negative Matrix Factorization (NMF)
18. BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies)

C. Reinforcement Learning Algorithms

01. Q-Learning
02. Deep Q-Networks (DQN)
03. Policy Gradients
04. Proximal Policy Optimization (PPO)
05. Trust Region Policy Optimization (TRPO)
06. Actor-Critic Methods
07. Monte Carlo Methods
08. Temporal Difference Learning
09. SARSA (State-Action-Reward-State-Action)
10. Deep Deterministic Policy Gradient (DDPG)
11. Asynchronous Actor-Critic Agents (A3C)
12. Soft Actor-Critic (SAC)
13. N-step Q-Learning
14. Hierarchical Reinforcement Learning (HRL)
15. C51 (Categorical DQN)
16. Hindsight Experience Replay (HER)
17. Distributional Reinforcement Learning
18. Double Q-Learning